Features in Development

Colosseum: additions to the Phase-1 design pack (docs 01–13)

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The Phase-1 pack designs the scored assessment loop: challenge authoring, attempts, AI judging with cited evidence, ratings that decay, and the company and valuator flows. This document lists the features being added on top of it. Both new features are **learning features**: they help students see and improve what they can do, and they **never change a score or a rating**.

1. At a glance

	Feature	What it adds	Builds on
A	**Skill Mastery** (game style)	A skill tree that lights up as students prove sub-skills with scored work, and fades when skills go stale	Docs 04, 05, 07
B	**Assumption Tracking** (cognitive modelling)	Finds the assumptions a student's answers keep relying on, shows the evidence, and targets practice at them	Docs 04, 05, 06, 07, 08
C	**Shared Foundations**	Consent, privacy, data and AI-service additions that both features need	Docs 01, 03, 08, 09

All features by title

ID	Title	In one line
A1	Skill Tree	About 54 sub-skill nodes across CSE, ECE and IT, taken from doc 07
A2	Sub-skill Tags on Rubrics	Each rubric criterion names the sub-skills it tests
A3	Evidence-earned Tiers	Nodes light up only from scored work, never from activity
A4	Mastery Fade and Refresh	Tiers fade at each family's decay rate; one new proving answer restores them
A5	Traceable Mastery	Every tier change cites the score and quote that earned it
A6	Sub-skill Estimates	A hidden estimate per node that picks useful practice
B1	Assumption Catalogue	A fixed, signed-off list of assumptions per family
B2	Exposure Mapping	Which assumptions each challenge variant can reveal
B3	Answer Tagging	Each answer marked held, guarded or stated, with a quote
B4	Assumption Patterns	Emerging, Pattern and Outgrown states over time
B5	Your Assumptions Page	The student's view of patterns, evidence and progress
B6	Targeted Practice	Practice variants chosen to expose open patterns
B7	Tag Disputes	"That's not what I assumed": hide, log and learn
B8	Cohort Assumption Insights	Aggregate patterns for colleges, cohorts of 25 or more
C1	Learning-profile Consent	A separate consent purpose, off by default
C2	Company Exclusion	Companies see none of this data, enforced in code
C3	Data Model Additions	Nine new tables, append-only where it matters
C4	Assumption-tagging Endpoint	One new AI-service endpoint with quote validation
C5	Quality Measurement	Precision, agreement, disputes and learning effect

2. Feature A: Skill Mastery (game style)

Shows students, skill by skill, what they have proven with real scored work, in a form that is fun to fill in and honest about skills going stale.

A1 Skill Tree

- One tree per domain, built from the doc 07 taxonomy: 18 challenge families with three sub-skills each, about 54 nodes. Example: Backend & APIs → REST/gRPC, Idempotency, SQL.
- Nodes start unlit. Nothing lights up for signing in, streaks or attempting.

A2 Sub-skill Tags on Rubrics

- When a template is authored (doc 05), each rubric criterion is tagged with the sub-skills it tests.
- The certifier checks that every certified variant can exercise each tagged sub-skill, extending the existing rule that every variant must exercise every criterion.
- Cheap to add now: one column in authoring, so evidence builds up from the first attempt.

A3 Evidence-earned Tiers

Tier (working name)	Earned by
Unlit	No qualifying evidence yet
Proven	Criterion level 3 or higher on one certified variant tagged with the sub-skill
Mastered	Level 3 or higher on two different certified variants
Needs a refresh	A tier whose evidence has faded (A4)

- Ranked and practice attempts both count. Tiers never feed the rating.
- Tier names are deliberately different from rating bands, so nobody reads a tier as a hiring score.
- Outgrowing an assumption (B4) can also light a node.

A4 Mastery Fade and Refresh

- Uses the same per-family decay rates and 90-day grace period as the rating (doc 04 §4, doc 07 §4).
- When the faded estimate drops below a tier's bar, the node shows **Needs a refresh**. One new qualifying answer restores it.

A5 Traceable Mastery

- Every tier change is an append-only event that cites the criterion score, and through it the quoted evidence, that earned it.
- Mastery can be rebuilt from evidence the same way a score can.

A6 Sub-skill Estimates

- Behind the tiers, each node keeps an estimate with an uncertainty, using doc 04's update rule at sub-skill level. It picks useful practice and is not shown as a number.
- No neural or item-response models until each node has thousands of observations, for the same reason doc 04 defers them.

Who sees it	What they see
Student	The full tree, tiers and the evidence behind each tier
College	Aggregates only, for cohorts of 25 or more
Company	Nothing in the first version

3. Feature B: Assumption Tracking (cognitive modelling)

Finds the assumptions a student's answers keep relying on that the challenge never guaranteed, shows the evidence, and targets practice at them.

The Phase-1 docs already contain many such assumptions:

- Doc 05 webhook challenge: each event is delivered once; events arrive in order; only one worker processes an event at a time.
- Doc 07 settlement worker: the balance I read is still current when I write.
- Doc 07 rubric levels: the result set is never empty; the network is trusted; the bus runs at the speed I tested.
- Doc 06 viva question: "What breaks if the queue is at-least-once instead of exactly-once?"

B1 Assumption Catalogue

- A fixed list, signed off by a practitioner and version-locked like rubrics (D2).
- About 20 shared assumptions (scale, input validity, time, concurrency, partial failure, data changing underneath you, silent failure, trust boundaries) plus about 10 per family.
- Each entry holds a plain statement, what the evidence looks like, the rubric levels it explains and the related sub-skills. Seeded from doc 07's rubric levels.

B2 Exposure Mapping

- Each template's variable values (doc 05 §3.1) are tagged with the assumptions they can expose. Example: `failure_mode = out_of_order` exposes "events arrive in order".
- Patterns are measured as **held ÷ exposed**, so a student who simply got more concurrency variants does not look worse at concurrency.
- The certifier checks that each tagged assumption can be exposed by that variant.

B3 Answer Tagging

Outcome	Meaning	Example
Held	The answer relied on it without saying so	Inserts the payment with no check for an earlier delivery
Guarded	The answer handled the case	Uses an event-derived key written in the same transaction
Stated	The answer named it as a deliberate choice	"I assume a single consumer; scaling out needs a lock"

- One small-model call per scored answer, ranked or practice. It returns tags, each with an outcome and an exact quote from the answer.
- The server refuses any tag whose quote is not in the answer or whose assumption was not exposed: the same check the judge's evidence gets.
- Kept separate from the judge, so changing the catalogue never forces a new judge version.
- Stating an assumption is good engineering and never counts against the student.

B4 Assumption Patterns

State	Rule (starting targets)	Visible to
Emerging	Held twice	Student only
Pattern	Held 3 or more times, in at least half the exposures, across at least 2 templates	Student; counted in cohort totals
Outgrown	Was a pattern, then guarded on each of the last 3 exposures	Student; can light a mastery node

B5 Your Assumptions Page

Illustrative example:

> **You tend to assume only one worker touches a record at a time.**
> Held in 3 of 5 answers that could show it, across 3 challenges.
> · Webhook challenge: the duplicate check and the insert run in separate transactions
> · Settlement worker: balance read and debit with no lock or version check
> **Try:** a practice variant with 16 concurrent workers.
> **Outgrown so far:** "the result set is never empty" ✓

- Wording describes the work, never the person: "your answer assumed", not "you are careless".
- On scorecards, tags can explain why a criterion lost a level, but they never change the score.

B6 Targeted Practice

- In practice mode, the variant is chosen to expose the student's open patterns, for example `workers = 16`.
- Ranked attempts keep the normal seeded draw, so targeting can never affect a rating.

B7 Tag Disputes

- Each tag has a "that's not what I assumed" button. A disputed tag is hidden, and the dispute is logged as a new row.
- The dispute rate per assumption is the main quality signal. Entries that keep getting disputed are rewritten or retired.

B8 Cohort Assumption Insights

- Colleges see totals for cohorts of 25 or more. Illustrative: "Top assumptions in CSE 2027: single worker (52%), result never empty (38%)."
- Curriculum feedback with no view of individual work (doc 01 §6).

4. Feature C: Shared Foundations

Additions both features depend on.

C1 Learning-profile Consent

- A new consent purpose, off by default, covering mastery and assumption data.
- The erasure job (doc 09 §1.1) covers every new table.

C2 Company Exclusion

- Mastery and assumption data never appear in company search or unlocked scorecards in the first version.
- Enforced in the disclosure code (doc 08 §1.3.1), not by page layout.

C3 Data Model Additions

Table	Purpose
`skill_node`	The ~54 sub-skills from doc 07
`criterion_skill_tag`	Which sub-skills each rubric criterion tests
`mastery_event`	Append-only tier changes, each citing a criterion score
`mastery_state`	Current tier and estimate per student and node
`assumption`	Catalogue entries; versioned, immutable once published
`item_exposure`	Which assumptions each variable value or variant can expose
`assumption_observation`	Append-only tags: outcome, quote position, classifier run
`assumption_dispute`	Disputes, inserted as new rows
`assumption_profile`	Current counts and state per student and assumption

- Students see only their own rows; college views enforce the 25-student minimum.
- A new top-level **mastery** module needs an ADR (doc 03 §5).

C4 Assumption-tagging Endpoint

- `POST /v1/tag-assumptions` on the AI service. It accepts only the assumptions the variant exposes and returns tags with quotes, never free text.
- Same safety rules as the existing endpoints: no database credentials, no storage writes (doc 03 §3).

C5 Quality Measurement

Measure	Target
Tag precision (a practitioner audits about 50 tags a week)	0.85 or higher before patterns are shown
Students agree a tag describes their answer	70% or higher
Dispute rate per assumption	Tracked; outliers rewritten or retired
Learning effect: held rate after targeted practice	Must fall, or the feature is not teaching

5. Rules both features keep

- They never change a score or a rating.
- They read only submitted work, never typing, paste or focus data.
- Every tier change and every tag is backed by quoted evidence.
- Fixed, signed-off vocabularies; no free-text labels about people.
- No points for participation.
- Companies see nothing in the first version; colleges see aggregates of 25 or more.
- Off by default, under their own consent purpose.

6. Cost

Item	Estimate
Skill Mastery	No model cost; computed from existing criterion scores
Assumption Tracking	About one small-model call per scored answer, roughly ₹0.25 (est., based on doc 03's integrity classifier price)
Practitioner time	Writing and signing off the catalogue; auditing about 50 tags a week

7. Build order

Step	What gets built	Moves on when
0 (now)	Sub-skill tags on criteria and exposure tags on variable values, added to template authoring	Every new template carries both tags
1	Skill tree and tiers from existing criterion scores	Tiers rebuild exactly from events in a replay test
2	Assumption catalogue for one browser-mode CSE family (Algorithms or Security); 50 answers hand-tagged	Catalogue signed off
3	Tagging classifier, offline	Precision 0.85 or higher against the hand tags
4	Per-answer tags on scorecards, with disputes	Student agreement 70% or higher
5	Patterns, Outgrown and targeted practice	Held rate falls after targeted practice
6	College cohort insights	A cohort of 25 or more is available

Step 0 can start now. Steps 1–6 come after the committed quarter's core loop (decision D14).

8. Open decisions

Decision	Recommendation
Tier names and thresholds	Keep the working names until student testing
Show assumption tags on rated scorecards, or only in practice?	Both; never in company views
Should targeted practice use a student's history?	Yes, in practice mode only
Who writes the catalogue?	The CSE subject reviewer (doc 13), signed off by the lead
Should mastery ever be shown to companies?	Not in the first version

9. Phase-1 docs these features touch

Doc	Change
01 System architecture	New mastery module alongside the existing twelve
03 Tech stack	ADR for the new module; one new AI-service endpoint
04 Scoring and decay	Update and decay rules reused at sub-skill level; the rating itself is unchanged
05 Authoring engine	Sub-skill tags and exposure tags; new certifier checks
06 Integrity	Confirms activity telemetry is not used by these features
07 Domains	Source of the skill tree and the catalogue seeds
08 Company and valuator flows	Disclosure code excludes the new data
09 Data model and API	Nine new tables and one endpoint